Designing a Hybrid Model for Chronic Respiratory Disease using Lung Sound Classification

Khushi Gupta
Department of Data Science,IoT and Cyber Security(DIC)
G H Raisoni College of Engineering
Nagpur,India
khushi.gupta.ds@ghrce.raisoni.net

Medhavi Nasare
Department of Data Science,IoT and Cyber Security(DIC)
G H Raisoni College of Engineering
Nagpur,India
medhavi.nasare.ds@ghrce.raisoni.net

Shilpa Tichkule
Department of Data Science,IoT and Cyber Security(DIC)
G H Raisoni College of Engineering
Nagpur,India
shilpa.tichkule.ds@ghrce.raisoni.net

Nekita Chavhan Morris
Head of Department
Department of Data Science,IoT and Cyber security (DIC)
G H Raisoni College of Engineering
Nagpur,India
nekita.chavan@raisoni.net

Rahul Agrawal
Assistant professor
Department of Data Science,IoT and Cyber security (DIC)
G H Raisoni College of Engineering
Nagpur,India
rahul.agrawal@raisoni.net

Chetan Dhule
Assistant professor
Department of Data Science,IoT and Cyber security (DIC)
G H Raisoni College of Engineering
Nagpur,India
chetan.dhule@raisoni.net

***Abstract*—Classification of lung sounds is one of the critical tasks in diagnosis for doctors and clinicians as it helps them in the diagnosis of several diseases in the respiratory system. Some of the chronic respiratory diseases, normally attributed to abnormal lung sounds, are clinically correlated with conditions such as bronchitis and chronic obstructive pulmonary disease. However, environmental noise mixed with abnormal lung sounds present challenges that require improved classification methods. In this paper, we focus on lung sound classification using the ICBHI 2017 dataset. We employ Mel-frequency cepstral coefficients (MFCC) for feature extraction. Additionally, we incorporate the SMOTE to address the problem of class imbalance and the hybrid model of SVC and KNN. Our work also includes a comparative analysis between the various traditional machine learning models in the context of lung sound classification. Our model gained an accuracy of 79.56%, demonstrating an improvement over the previous related work. This comprehensive approach aims to capture lung sound data more effectively, differentiating between various types of lungs sounds and providing improved diagnostic assistance to healthcare professionals.**

Keywords—Support vector classifier, KNN, Hybrid, SMOTE, Mel-frequency cepstral coefficients

I. INTRODUCTION

Lung sounds, sometimes referred to as breath sounds or respiratory sounds, are the natural sounds created by air traveling effectively through the respiratory system, specifically the lungs, during breathing. There is usually listening to the normal and abnormal sounds with a stethoscope in auscultation and important evidence of health regarding lungs and possible respiratory illness. There are three major types of lung sounds. First is the Normal Breath Sounds which include the vesicular sound, bronchial sounds. Second is an Abnormal Sounds which includes crackles, stridor, rhonchi and wheezes. And the third one is the decreased or absent breath sound. Chronic Respiratory Disease, or CRD for short, is one of the world's major chronic conditions in which there is a decline in respiratory functions because of various continuous non-contagious diseases. According to a report by the WHO, it is estimated that 400,000 people die every year due to chronic respiratory disease. Of these deaths, a large amount stems from Chronic Obstructive Pulmonary Disease which is a progressive lung disease, essentially characterized by shortness of breath, labored breathing, coughing, and overproduction of mucus. Advanced states of the disease significantly affect the quality of life and working ability of patients, sometimes even leading to disability. More than 300 million patients in the world suffer from COPD; that number is expected to increase by 2030[2]. India will face the third largest economic burden due to COPD between 2020-2050 after China and the US, says recent research published in the journal, The Lancet Global Health. In 2019 alone, as many as 3·3 million died due to COPD; it is considered the third leading cause of deaths globally. Among these, China had the maximum number of deaths, followed by India and the USA in ranking order. Some diagnostic methods performed by healthcare professionals include:[1]

1. Pulmonary Function Tests: These examinations help to detect the effectiveness of the lungs and if they are managing to provide adequate oxygen during respiration. The most common test is through spirometry, where the patients breathe in and out into the spirometer, which records volume and airflow.

2. X-ray of the chest: It allows such imaging to visualize diseases of the lung, including emphysema, considered one of the major causes of COPD.

3. CT scan: Not in common use, but CT scan is employed in advanced stages of COPD where the conventionality of diagnostic tools is not helpful. It also helps in determining whether surgical intervention for COPD is required or not.

4. Arterial Blood Gas: This test is a way to find out just how well oxygen is getting through from lungs into the blood-essentially, it measures oxygen levels in the blood.

Limitations of Current Diagnostic Techniques [1]

1. This test is not suitable for patients with any heart condition or patients who have undergone recent heart surgery. It may not yield appropriate results or could even worsen their health issues.
2. Radiations associated with CT scans and X Ray are hazardous to the health of patients, especially when exposures are repeated.
3. Tests one undergoes can further cause some uncomfortable symptoms and anxiety of breathlessness, nausea, and dizziness.

Recent times have seen phenomenal advancements in deep learning and machine learning. These are expanding the horizons of medical research by opening new vistas for early detection and classification of COPD. These techniques automatically identify the pattern and predict them from volumes of data. This might allow the application of various machine learning approaches in respiratory sound data analysis to assist the clinician in making a proper diagnosis and categorization of patients. Machine learning, if there are big datasets of respiratory sounds for training and evaluation of patients, can classify and identify such sounds rather systematically than subjective assessment by the healthcare professional.

Nevertheless, this research was conducted to classify respiratory sounds based on the feature extraction technique of MFCC. As a matter of fact, MFCC has tended to become a standard feature extraction technique for processing audio signals, particularly in speech and sound classification. Our proposed model, which reflects the use of the hybrid model, combines Support Vector Machine and K-Nearest Neighbors. While SVM gives a robust decision boundary, KNN provides the flexibility for local decision-making. A hybrid model avails the advantages of both models to offer better accuracy and generalization on the Respiratory Sound dataset.

II. LITERATURE SURVEY

Srivastava et al. 2021[1] put forward a Convolutional Neural Network approach toward respiratory sound analysis in characteristic features like MFCC and Mel-Spectrograms to find out the acuteness of COPD. Using this, the research was able to come across that audio analysis with K-Fold Cross-Validation produces higher results regarding model performance and classification accuracy.[2] used an advanced Bi-ResNet network model for the purpose of enhancing classification precision by incorporating skip connections and direct feature transmission. By combining convolutional neural networks (CNN) with residual networks and using both wavelets transform and STFT features, the model improves the evaluation of lung sounds. Resulting in 77.81% classification accuracy on ICBHI 2017 dataset. [3] They classify respiratory sounds, focusing on crackles and wheezes using machine learning, especially using the ICBHI 2017 dataset. The initial model, boosted decision tree and audio features, achieved around 50 % accuracy. To improve performance, a new model that evaluates predictions at the patient level rather than on individual respiratory cycles was introduced. This approach improved accuracy to 85%. [4] focuses on using computer-based analysis to identify crackles in lung sounds. The approach involves recording lung sounds with a respiratory sensor and microphone, then applying continuous wavelet transform to remove noise. Spectrograms and scalograms are generated from the transformed signals, and features are extracted to differentiate between normal and adventitious lung sounds. [5] proposed deep learning models applied to electrocardiogram (ECG) signals for diagnosing COPD. They used CNN models such as Xception and VGG-19, with the Xception model achieving 99% accuracy.[6] study focuses on developing an algorithm for the classification of respiratory sounds using the ICBHI 2017 Challenge dataset. One-dimensional respiratory sound signals transformed into two-dimensional images through STFT and continuous wavelet transform. These images were then classified using CNN. The proposed method was applied to sound recordings, achieving a score of 28%, an 81% of harmonic score, a 54% sensitivity, and a 42 % of specificity. Jindal et al. [7] used the KNN machine classification method to integrate the burst tone parameters to the vector space, which enhanced detection accuracy. Earlier studies based on lung sound classification include machine learning methods, probabilistic statistical classification methods, and support vector machine (SVM)-based lung sound classification. Bahoura and Pelletier [8] discovered that highest sensitivity is achieved by combining MFCC and GMM to classify the lung sounds into two main categories normal and wheeze. Abbasi et al. [9] proposed the use of SVMs to classify normal and adventitious lung sounds which achieved the classification performance better than feed-forward neural networks (NN) and probabilistic NN. Jan Feiba and colleagues [10] have elaborated on classification of pathological lung sounds and sources of noise interference of lung sound signals by the lung sound monitoring system, they have brought to light the existence of current methodologies of feature extraction comprising: STFT, wavelet analysis identification and higher-order spectral analysis for analyzing and detecting pathological lung sounds. [11] proposed that changes in lung sounds can effectively reflect changes in lung structure associated with diseases. Resulting in classification accuracy of 99.7% by using Empirical Mode Decomposition (EMD) and Discrete Wavelet Transform (DWT) to clean the recordings. [12] The authors have performed the classification from the images of spectrograms using a CNN model-based parallel-pooling structure. Then, using the fully-connected layers of CNN, features were extracted and further classified by LDA combined with the RSE method. These approaches have achieved an accuracy of 71.15% for the ICBHI 2017 dataset.[13] Hybrid novel CNN-LSTM is used. CNN extracts feature from spectrogram images of lung sounds from ICBHI 2017 dataset. These features are then processed by LSTM. For imbalanced data, the model incorporates focal loss (FL) technique. Based on three data splitting methods, the model produced different outcomes. For the 60/40 split, it reached a sensitivity of 47.37%, a specificity of 82.46%, a score of 64.92%, and an accuracy of 73.69%. When interpatient 10-fold cross-validation was applied, the sensitivity increased to 52.78%, specificity to 84.26%, the score to 68.52%, and accuracy to 76.39%. Lastly, with leave-one-out cross-validation (LOOCV), the model achieved a sensitivity of 60.29% and an accuracy of 74.57%.[14] The study developed a machine learning model using ensemble support vector machines to classify pediatric respiratory sounds recorded with electronic stethoscopes. Digital recordings of lung sounds from routine physical examinations in a pediatric pulmonology outpatient clinic were labeled as crackles , normal , or wheezing and used to train and evaluate ensemble support vector machine models for the following four classification tasks: normal versus abnormal, crackles versus wheezing, normal versus crackles, and normal versus wheezing, using K-fold cross-validation (K = 10).The model was trained on 680 clips and evaluated on 90 additional clips, achieving high accuracies in internal validation (up to 90.42%) and comparable performance to

pediatricians and non-pediatricians. [15] The ResNet models have been applied for the classification of adventitious lung sound classification and disease diagnosis based on ICBHI and multi-channel lung sound datasets. Transfer of the learned representations is performed by vanilla fine-tuning, co-tuning, stochastic normalization, and their combinations. Spectrum correction and data augmentation by flipping are applied to enhance the robustness of the system. These proposed systems gave an average score of 58.29% and 64.74% for 4-and 2-class lung sound tasks and an average score of 92.72% and 93.77% for 3- class and 2-class respiratory disease tasks, respectively.

Table 1: Result of the previous work using different methods on the same dataset.

Papers	Models	Dataset used	Accuracy	Year
Bi-ResNet Network for Enhancing Classification Precision	Bi-ResNet, STFT, Wavelet Transform	ICBHI 2017	77.81%	2024
Bi-ResNet Network for Enhancing Classification Precision	CNN	ICBHI 2017, Mendeley Data	91.04% (10-classes)	2024
AusculNET: DL framework for Adventitious Lung Sounds Classification	CNN (3 layers), LSTM (1 layer), Log-Mel Spectrogram	ICBHI 2017	Accuracy: 58%, ICBHI Score: 55%	2023
CNN-Based Parallel-Pooling Structure for Lung Sound Classification	CNN, LDA with RSE	ICBHI 2017	71.15%	2020
Co-Tuning and Stochastic Normalization for Classification	ResNet	ICBHI 2017	58.29% (4-class lung sound), 64.74% (2-class lung sound), 92.72% (3-class respiratory disease), 93.77% (2-class respiratory disease)	2020
Automatic Classification using CNN	CNN, STFT, Wavelet Transform	ICBHI 2017	28% (Score), 81% (Harmonic), 54% (Sensitivity), 42% (Specificity)	2019
Patient with Respiratory Diseases Automatic detection	Boosted Decision Tree, Patient-Level Predictions	ICBHI 2017	Initial: 50%, Improved: 85%	2018

III. METHODOLOGY

The proposed methodology aims to address the gaps identified in the current research on classification of lung sounds and to improve the model's performance. The methodology integrates solutions for machine learning-based respiratory sound analysis using hybrid models. Utilize Mel-frequency cepstral coefficients (MFCC) and other time-frequency representations to ensure comprehensive feature capture, Synthetic Minority Oversampling Technique (SMOTE) generating synthetic sample for minority classes to better balance the dataset and Explore hybrid models that combine traditional machine learning methods to leverage the strengths of both SVM and KNN as shown in figure 1.

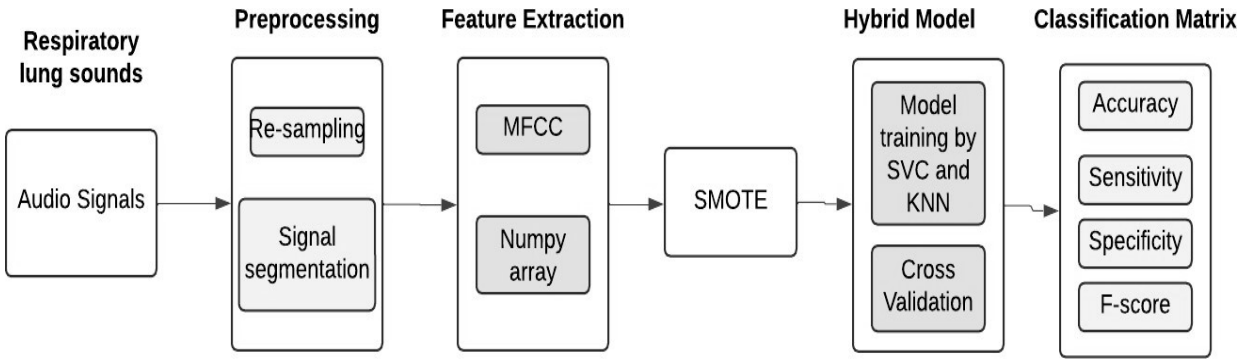


Fig. 1. Workflow of the proposed model.

A. Dataset Preparation

In this study, the dataset used is the ICBHI 2017 dataset. The data include 920 audio recordings with a recorded duration range between 10 to 90 seconds. The dataset was obtained from the recordings of a total of 126 participants in the study. Table 2 demonstrates that in total, approximately 5.5 hours of recordings have been recorded, and contained 6898 respiratory cycles. To reflect real world conditions the dataset includes both clear respiratory sounds and recordings with background noise.

Table 2: ICBHI 2017 dataset Cycle breakdown

Number of Cycles	Total
Normal Cycles	3642
crackles	1864
Wheezes	886
wheezes + Crackles	506
Total number of cycles	6898

B. Preprocessing

The ICBHI 2017 dataset contains two abnormal lung sounds: crackles and wheezes. The crackles are low-pitched pulses lasting less than 20ms, and usually occur in rapid succession. These sounds are often associated with bronchitis, COPD and many other conditions. Whereas, wheezes are high-pitched, have musical character and last longer than 250 ms. Obstructive pulmonary disease is often linked to diseases of the lungs, for example, asthma and cystic fibrosis.

In Preprocessing, the .wav files were first extracted from the dataset directory. A Data Frame is created to sort sound files by patient ID, which is assumed to be the first three characters of each filename. This allows audio files to be formatted in a structured and patient-specific manner. Each audio file is loaded and its sampling rate is checked. If the sample rate of a file does not match the target rate i.e, 22,000Hz, it is resampled using linear interpolation. This ensures consistency across recordings, as different sample sizes can lead to differences in feature extraction analyses.

The audio data is then processed to extract logical segments based on the annotations in the corresponding .txt files. These annotations consist of start and end times of the respiratory cycles which are converted to sampling indices by multiplying the time (in seconds) by the sampling rate (22,050 samples per second) This allows complete records to be segmented for short breathing cycles. Then the Labels are generated based on crackles and/or wheezes of each respiratory cycle. Where 0 is Normal, 1 is Crackles, 2 is Wheezes and 3 is for both crackles and wheezes. Each breathing cycle is then segmented from audio recordings based on the start and end times. And corresponding label is assigned to each segment based on annotated data.

The final segmented audio data and their corresponding labels are prepared for feature extraction (e.g., MFCC) and model training.

C. Feature Extraction

In the feature extraction phase, This research focuses on the use of Mel-frequency cepstral multipliers (MFCCs), which are widely recognized for their effectiveness in representing audio signals, especially for speech and biomedical sound analysis. MFCCs convert the raw audio signal into a frequency domain, specifically targeting human auditory perception by emphasizing the lower frequencies. This therefore, makes MFCCs most helpful in analyzing respiratory sounds since they generally contain low-frequency components such as wheezes and crackles.

We applied MFCC extraction to the audio recordings with specific parameters to maintain temporal and spectral consistency. For the sample rate, a window frame size of 25ms was chosen, representing 25% of the sample rate, while the hop frame shift was taken as 10ms. These ensure sufficient overlap between successive windows so that the fine temporal details of the sounds, such as the short bursts of crackles or the high-pitched wheezes which may be elongated, are captured.

From the MFCCs, we computed two important statistical features, namely, the mean and standard deviation. These statistics describe general properties in time of the distribution of the values of the MFCCs for each sample - providing both the mean representation and variability through the standard deviation of the sound characteristics.The feature vectors are organized into a numpy 2D array, that put together these vectors in such a way that each of the row corresponds to an audio segment and each column represents a different feature. By reducing the sequence of MFCCs to these statistics, we obtain a compact feature set which retains the essential information necessary for classifying respiratory sounds and minimizes computational complexity.

D. Evaluation Methodology

The audio segments are combined with the extracted MFCC features to create a feature vector for each one. In this vector, factors of the MFCC coefficients are incorporated in order to provide a brief account of audio's spectral attributes. A numpy 2D array is where these feature vectors are put in order with respect to audio segments as their rows and specific features as columns. By converting them into single integer values, the one-hot encoded vectors become labels related to every segment separately. As such, changing from one-hot encoded format to streamlining it helps provide a better fit for classification algorithms. It

simply means that there are numerical indices associated with these labels which represent different types of respiratory events like crackles or wheezes.The final dataset comprises both the feature vectors and their respective labels which have been rearranged and formatted for training the models. The feature matrix is structured ensuring that each row corresponds to an audio segment while all columns represent different features obtained from the audio data. A well-structured dataset serves as an important factor when it comes to training and evaluating classification models accurately.

Currently, the data is segregated into a training and testing portion in the 80:20 split . A portion of the dataset called training data consists of MFCC (Mel-Frequency Cepstral Coefficients) features derived from audio signals used to train the models. These features represent the spectral characteristics of sound and are appropriate for machine learning models. The training labels, y_train, correspond to specific respiratory sounds or conditions related with every segment of audio. Prior to training, standardization was done using StandardScaler on train features ensuring that each feature has a mean equal to 0 while its standard deviation equals 1 thus making models converge effectively and operate optimally as well .After splitting the dataset into training and test sets, SMOTE is then applied to resample it, which is used to balance the dataset by generating synthetic samples so that model doesn't become biased toward the majority class .Once the data is rescaled using StandardScaler to standardize the features, several machine learning models such as SVM ,Gradient Boosting, Random Forest, KNN, Naive Bayes, XGBoost, and Decision Tree are trained on the balanced training data. The trained models are then evaluated using the performance metrics like accuracy, precision, sensitivity, F1-score, and confusion matrices. Cross-validation is also performed to get a more robust estimate of model performance across different folds of the training set. From the various machine learning models which has been trained the SVC model was particularly fine-tuned with hyperparameters and grid search such as class weight set to 'balanced', gamma value of 0.03125, and a regularization parameter C of 4, which were determined through prior optimization efforts and the KNN model was particularly fine-tuned with hyperparameters such as metric set to manhattan , n_neighbors as 3 and weight as distance. From these performance metrics we found that the SVC and KNN showed the higher accuracy score as compared to the other models that were evaluated. Additionally, a hybrid model was created by combining the predictions from the SVM and KNN classifiers. The predicted probabilities from these models were concatenated and used as features for a Logistic Regression meta-model, aiming to leverage the strengths of each base model for improved classification accuracy.

IV. EXPERIMENTAL WORK/ RESULT

A. Evaluation method and criteria

In our dataset , 80% of the data was used for training and the remaining 20% was used for the testing purpose. This both the training and the testing data is being extracted from the Mel-Frequency Cepstral Coefficients (MFCC) which is being stored in the form of numerical representation of the audio signal . Overall numerical data that is used in classifying 4 classes .

The performance metrics criteria were chosen as follows:

1. Accuracy: It quantifies the number of accurately classified instances, which could be both positive or negative, relative to the total instances.
2. Specificity: It computes the ratio of real negatives that are correctly identified. It is also referred to as the True Negative Rate.
3. Sensitivity (r e c a l l): This calculates the actual positives that are correctly identified. It indicates how good the model can detect positive instances
4. Precision: It measures the ratio of predicted to be positive among the actual positives. It reflects how many of the positive predictions are correct.
5. The F-score : It is the harmonic mean between precision and recall. It combines both of them into a single number, which can come in handy in dealing with imbalanced datasets.

These metrics help calculate the performance of classification models.

B. Experimental setup and Result

The dataset employed in this research exploited several classes whose distribution was not equal, hence it was imbalanced. In this case, a test size of 20% meant that out of all the data collected from the study; 80% was used as training while 20% remained for testing purposes. To ensure outcome repeatability, random_state parameter was fixed at 42. Stratification using stratify parameters helped maintain class proportion in both training and test assignments. Such measures prevented biased estimations. The experimental work involved training as well as evaluating multiple models of machine learning to identify the most effective approach for the classification task. The models included Support Vector Classifier (SVC) with an RBF kernel, Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN), Naive Bayes, XGBoost, and Decision Tree. The SVC model was particularly fine-tuned with hyperparameters and grid search such as class weight set to 'balanced', gamma value of 0.03125, and a regularization parameter C of 4, which were determined through prior optimization efforts and the KNN model was particularly fine -tuned with hyperparameters such as metric set to manhattan , n_neighbors as 3 and weight as distance . After training, the models were evaluated using various performance metrics which provided insights into the models' predictive capabilities across different classes. From these performance metrics we found that the SVC and KNN showed the higher accuracy score as compared to the other models that were evaluated as shown in the Table 3. From the confusion matrix and other performance metrics we detect the class imbalance problem. We solve this problem by using SMOTE. SMOTE is a technique that creates synthetic samples for the minority classes, which balances out the class distribution in the training dataset. As a result, models can learn from all classes properly. Additionally, a

hybrid model was created by combining the predictions from the SVM and KNN classifiers. The predicted probabilities from these models were concatenated and used as features for a Logistic Regression meta-model, aiming to leverage the strengths of each base model for improved classification accuracy.

Table 3: Accuracy score for other classifiers

Models	Accuracy
SVC	75.1%
Random Forest	70.3%
KNN	76%
Naïve Baiyes	43%
XGBoost	74.2%
Decision Tree	54.9%
Gradient Boosting	64.8%

V. RESULT

Experimental results obtained from the proposed hybrid model attained a classification accuracy score of around 79.56 % on the test set, improved by 1.75% compared with previous work. It had also attained an F1 score of 74 %. Figure 2 displays the confusion matrix of the hybrid model graphically. Here, one can easily see the comprehensive metrics of each class. The "normal" class was classified with a high degree of precision, that is, 0.82, meaning there were very few false positives. A recall of about 0.76 was seen for the "crackles" class, which indicates good performance by the model in terms of true positive detection.

For this model, the macro-average specificity was 0.91, documenting a high rate at which it correctly identified negative instances across all classes-a broader assessment of the system's ability to discern correctly non-target classes overall, the result validates that the hybrid model is applicable for solving the multi-class classification problem as illustrated in Figure 3. A big factor that contributed to improving this model performance was the use of SMOTE for class balancing along with a hybrid approach.

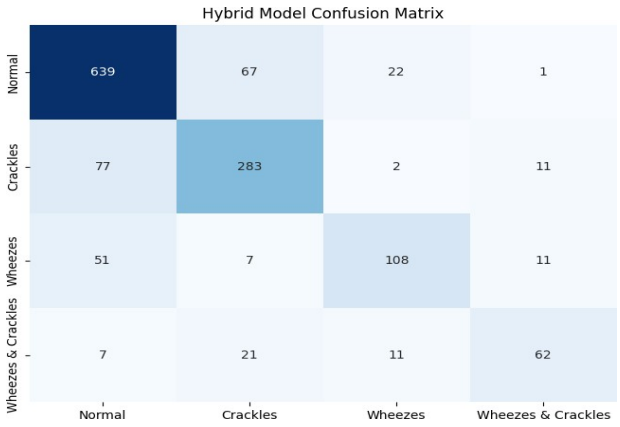


Fig.2 Confusion matrix for the proposed model.

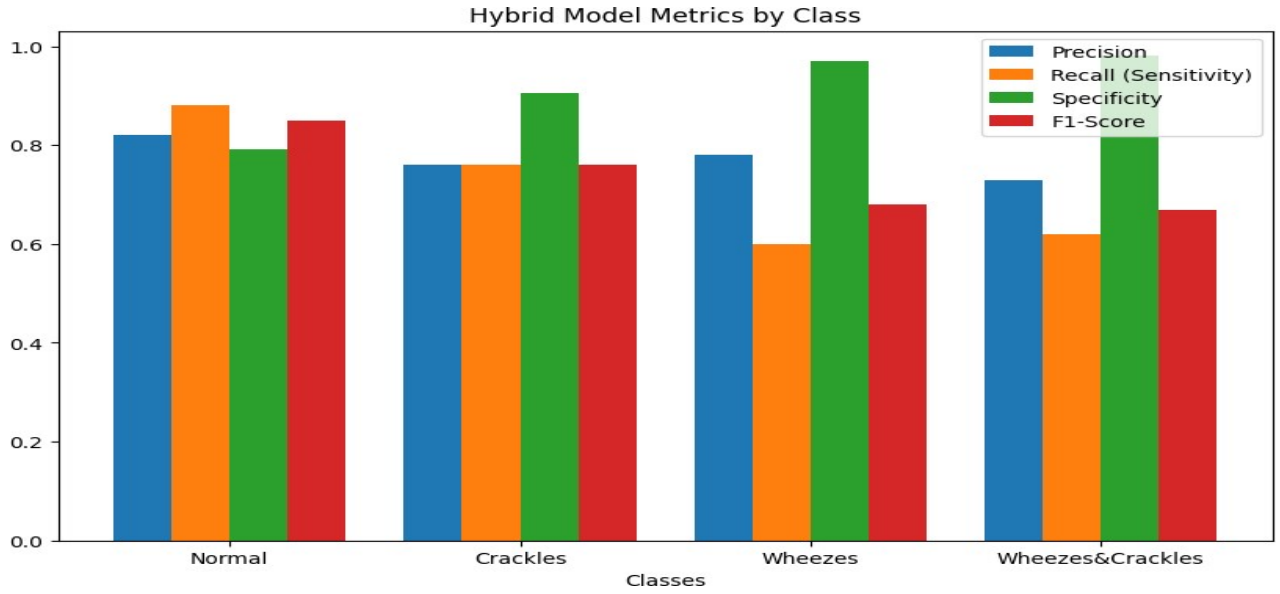


Figure 3: Result of hybrid model matrix by classes.

VI. CONCLUSION

These machine learning techniques were applied in this work for classifying the dataset ICBHI 2017, which contains respiratory sounds into the 4-classes.

Here, MFCCs were used to apply feature extraction and these managed to extract the significant features that define the respiratory sounds. Preprocessing such as resampling, segmentation and labeling helped to prepare the dataset systematically. We experimented with different machine learning models like SVM, KNN, Random Forest among others while SMOTE was used to balance the dataset in order to cater for class imbalance. Among all tested models SVM and KNN achieved better accuracy than the rest ones. Grid search for hyperparameters tuning led significantly towards improving their performances. Besides this hybrid model consisting of SVM and KNN was developed accompanied by a logistic regression meta-model to enhance further classification accuracy. The evaluation metrics under which these models were assessed included: accuracy, precision rate, recall rate as well as F1-score with cross-validation being used for ensuring robustness of results. Our experiments have shown that such methods can effectively work along with providing insights about advantages and disadvantages of different approaches. However, Future work could build on this by investigating sophisticated models, extra features, and stronger validation methods for improved efficiency and versatility in health care environments.

REFERENCES

[1] Srivastava, A., Jain, S., Miranda, R., Patil, S., Pandya, S., & Kotecha, K. (2021). Deep learning based respiratory sound analysis for detection of chronic obstructive pulmonary disease. PeerJ Computer Science, 7, e369.

[2] C. Wu, N. Ye and J. Jiang, "Classification and Recognition of Lung Sounds Based on Improved Bi-ResNet Model," in IEEE Access, vol. 12, pp. 73079-73094, 2024, doi: 10.1109/ACCESS.2024.3404657

[3] G. Chambres, P. Hanna and M. Desainte-Catherine, "Automatic Detection of Patient with Respiratory Diseases Using Lung Sound Analysis," 2018 International Conference on Content-Based Multimedia Indexing (CBMI), La Rochelle, France, 2018, pp. 1-6, doi: 10.1109/CBMI.2018.8516489.

[4] G. Shanthakumari and E. Priya, "Spectrogram-based detection of crackles from lung sounds," 2022 International Conference on Communication, Computing and Internet of Things (IC3IoT), Chennai, India, 2022, pp. 1-6, doi:10.1109/IC3IOT53935.2022.9768007.

[5] Moran, I., Altilar, D. T., Ucar, M. K., Bilgin, C., & Bozkurt, M. R. (2023). Deep transfer learning for chronic obstructive pulmonary disease detection utilizing electrocardiogram signals. IEEE Access, 11, 40629-40644.

[6] K. Minami, H. Lu, H. Kim, S. Mabu, Y. Hirano and S. Kido, "Automatic Classification of Large-Scale Respiratory Sound Dataset Based on Convolutional Neural Network," 2019 19th International Conference on Control, Automation and Systems (ICCAS), Jeju, Korea (South), 2019, pp. 804-807, doi: 10.23919/ICCAS47443.2019.8971689.

[7] Jindal, V., Agarwal, V., & Kalaivani, S. (2018, December). Respiratory sound analysis for detection of pulmonary diseases. In 2018 IEEE Applied Signal Processing Conference (ASPCON) (pp. 293-296). IEEE.

[8] M. Bahoura and C. Pelletier, "Respiratory sounds classification using cepstral analysis and Gaussian mixture models," The 26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, San Francisco, CA, USA, 2004, pp. 9-12, doi: 10.1109/IEMBS.2004.1403077.

[9] Shoichi Matsunaga, K. Yamauchi, Masaru Yamashita and Sueharu Miyahara, "Classification between normal and abnormal respiratory sounds based on maximum likelihood approach," 2009 IEEE International Conference on Acoustics, Speech and Signal Processing, Taipei, Taiwan, 2009, pp. 517-520, doi: 10.1109/ICASSP.2009.4959634.

[10] Jang, F. B., Yin, J., & He, Q. H. . Analysis and recognition method for pathological lung sound signal. Chin. J. Med. Phys, 33(7), 739-742.

[11] Haider, N. S., & Behera, A. K. (2022, December). A Brief Overview of the Respiratory Sound Based Detection of Chronic Obstructive Pulmonary Disease (COPD). In 2022 4th International Conference on Circuits, Control, Communication and Computing (I4C) (pp. 202-207). IEEE.

[12] F. Demir, A. M. Ismael and A. Sengur, Classification of Lung Sounds With CNN Model Using Parallel Pooling Structure, in IEEE Access, vol. 8, pp. 105376-105383, 2020.

[13] Petmezas G, Cheimariotis GA, Stefanopoulos L, Rocha B, Paiva RP, Katsaggelos AK, Maglaveras N. Automated Lung Sound Classification Using a Hybrid CNN-LSTM Network and Focal Loss Function. Sensors (Basel). 2022 Feb 6;22(3):1232. doi: 10.3390/s22031232. PMID: 35161977; PMCID: PMC8838187.

[14] Park, J.S., Kim, K., Kim, J.H. et al. A machine learning approach to the development and prospective evaluation of a pediatric lung sound classification model. Sci Rep 13, 1289 (2023).

[15] T. Nguyen and F. Pernkopf, "Lung Sound Classification Using Co Tuning and Stochastic Normalization," in IEEE Transactions on Biomedical Engineering, vol. 69, no. 9, pp. 2872-2882, Sept. 2022, doi:10.1109/TBME.2022.3156293.